Assignment No	6
Title	Ordinary Queue
Objective	Queues Operations
Roll No	MCA2649

Source Code:-

```
import java.util.*;
class ArratQueue
{
	//instance variables
	private int[] data;
	private int front;
	private int rear;
	private int capacity;

	ArratQueue(int size)
	{
		capacity=size;
		data = new int[capacity];
		front=-1;
		rear=-1;
	}
```

```
        //add method to enque
        public void enqueue(int ele)
        {
                // 1) Check whether the queue is full
                if(rear==capacity-1)
                {
                        System.out.println("Element can not be added! Queue is
full!");
                        return;
                }
                // 2) add first element in the queue
                if(front==-1)
                {
                        front=0;
                        rear=0;
                }
                else // 3) Any other element
                {
                        rear++;
                }
                data[rear]=ele; //4) add the data
                System.out.println("Element Added");

        }
        // Method to dequeue the data
        public void dequeue()
```

```
{
        // 1) is the queue empty
        if(front==-1)
        {
        System.out.println("The Queue is Empty!");
        return;
        }

        // 2) fetch and display data
        int x=data[front];
        System.out.println("Element Removed: " + x);

        // 3) for ant other element
        if(front==rear)
        {
        front = -1;
        rear = -1;
        }
        else
        {
        front++;
        }
}

// peek front
```

```
public void peekFront()
{
        if(front==-1)
        {
                System.out.println("The Queue is Empty!");
        return;
        }
        System.out.println("Front of the Queue:" + data[front]);
}

// peek rear
public void peekRear()
{
        if(rear==-1)
        {
                System.out.println("The Queue is Empty!");
        return;
        }
        System.out.println("Rear of the Queue:" + data[rear]);
}


void display()
{
        if(front==-1)
```

```
            {
                System.out.println("Queue is Empty!");
            }
            else
            {
                System.out.println("Elements in Queue!");

                for(int i=front;i<=rear;i++)
                {
                    System.out.println(" " + data[i]);
                }
                System.out.println();
            }
        }
}
class  QueueImpl
{
        public static void main(String args[])
        {
            ArratQueue a1=new ArratQueue(5);
            Scanner sc = new Scanner(System.in);
            int ch;
            do
            {
                System.out.println("Array Based Queue");
```

```
System.out.println("1-Enqueue an element");
System.out.println("2-Dequeue an element");
System.out.println("3-Display the Queue");
System.out.println("4-Peek Front");
System.out.println("5-Peek Rear");
System.out.println("6-Exit");

System.out.println("Enter your choice:");
ch=sc.nextInt();
switch(ch)
{
        case 1:
                System.out.println("Enter the element");
                int ele=sc.nextInt();
                a1.enqueue(ele);
                break;
        case 2:
                a1.dequeue();
                break;
        case 3:
                a1.display();
                break;
        case 4:
                a1.peekFront();
                break;
```

```
                    case 5:
                        a1.peekRear();
                        break;
                    case 6:
                        System.out.println("Exit");
                        break;
                    default:
                        System.out.println("incorrect choice");
                        break;
                }
            }while(ch!=6);

        }

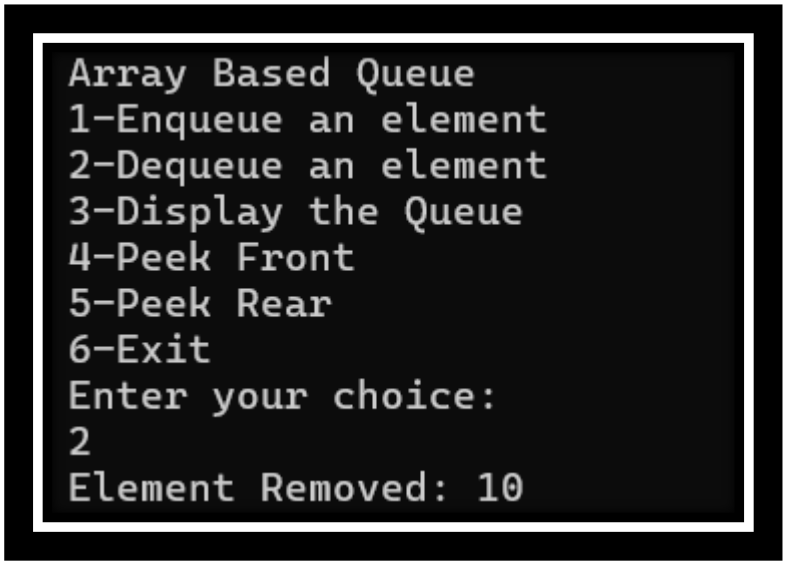
}
```

Output:-

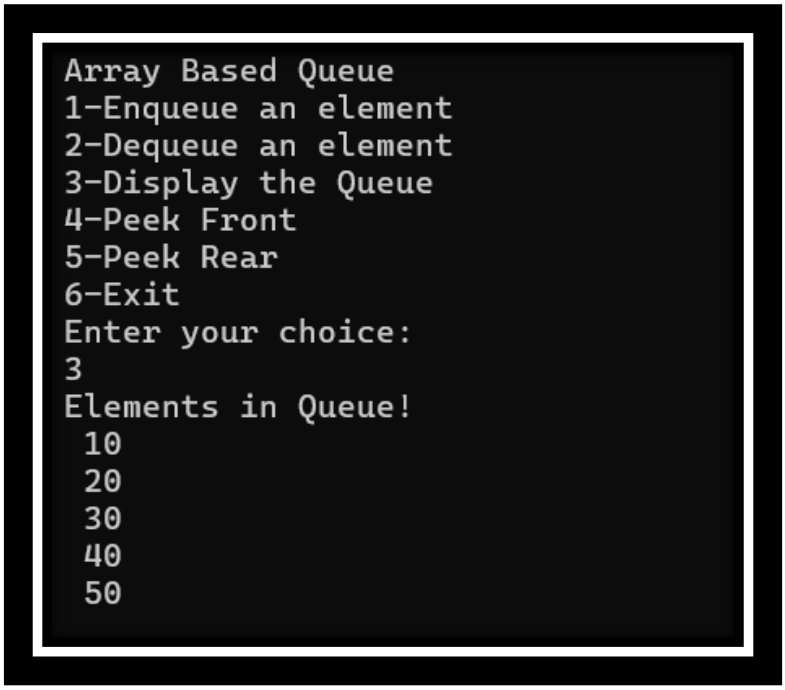
1) Enqueue an element

```
D:\MCA2649\ADS2649>java QueueImpl
Array Based Queue
1-Enqueue an element
2-Dequeue an element
3-Display the Queue
4-Peek Front
5-Peek Rear
6-Exit
Enter your choice:
1
Enter the element
10
Element Added
```

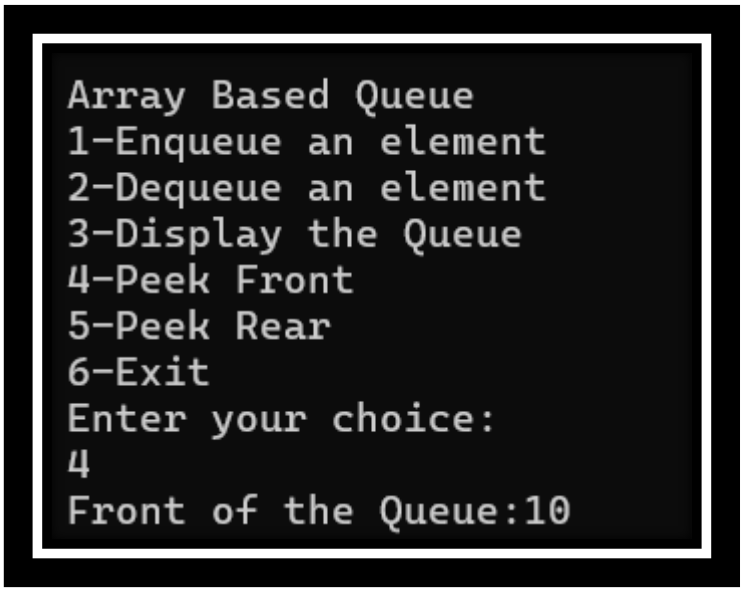
2) Dequeue an element



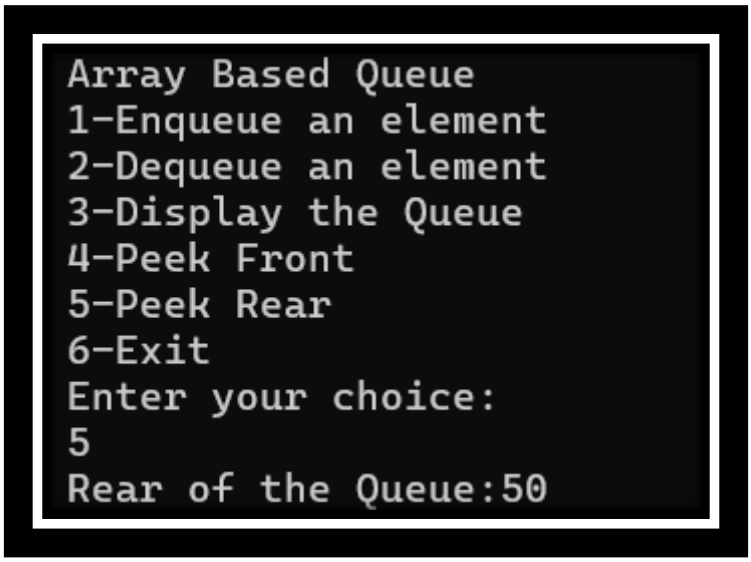
3) Display the Queue



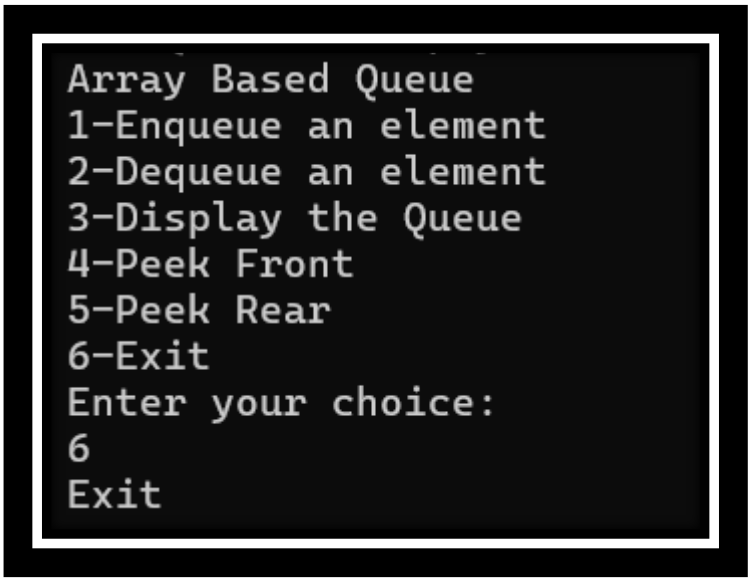
4) Peek Front



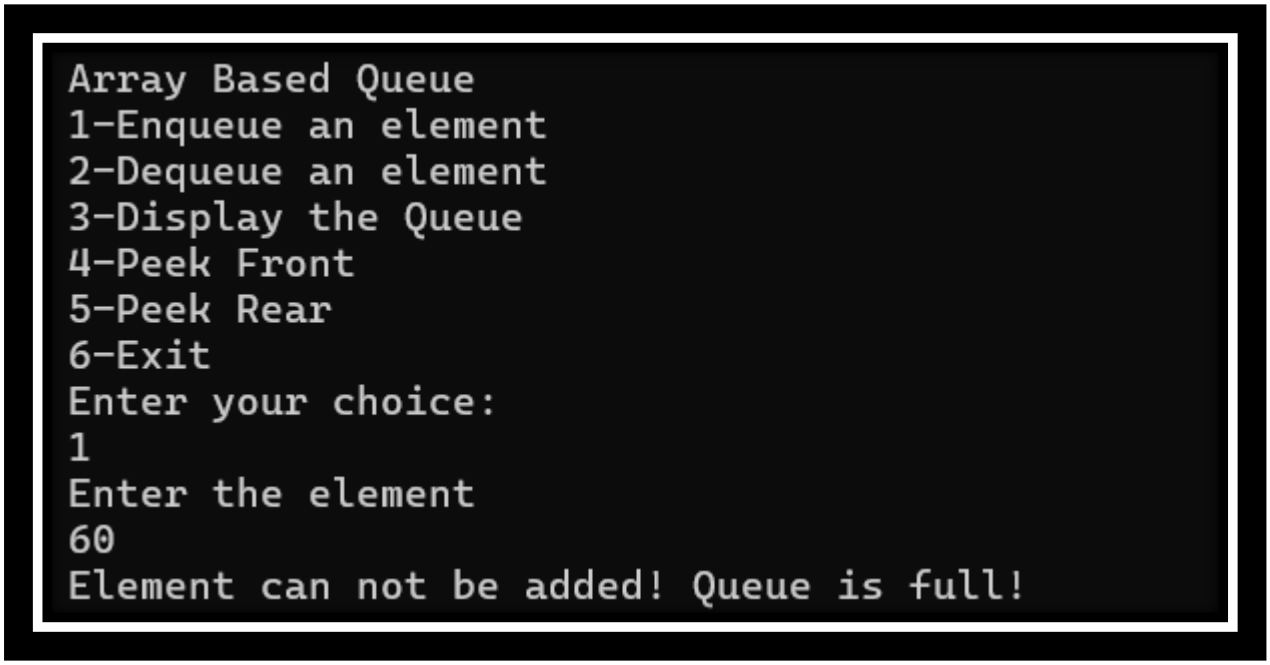
5) Peek Rear



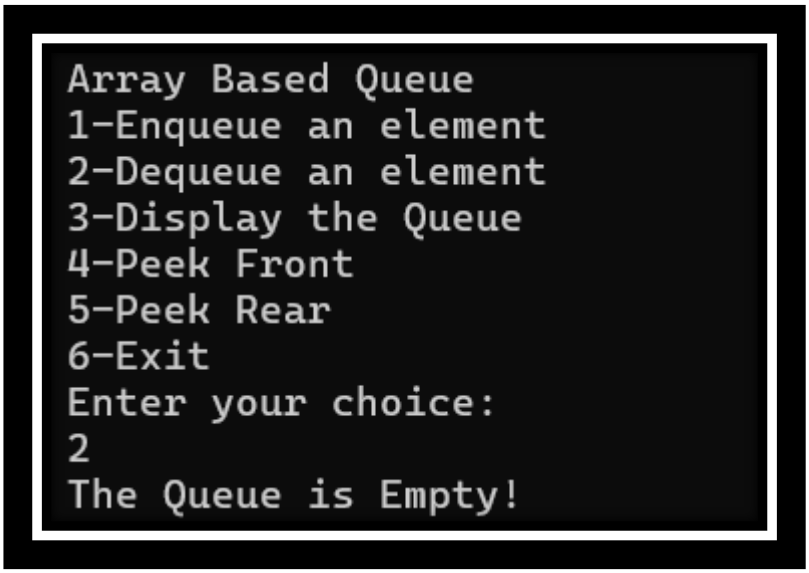
6) Exit



7) Queue is full!



8) Queue is Empty!


Array Based Queue
1-Enqueue an element
2-Dequeue an element
3-Display the Queue
4-Peek Front
5-Peek Rear
6-Exit
Enter your choice:
2
The Queue is Empty!